Strategy	Description	Examples	Revenue impact
Direct monetisation	Selling/licensing data to external users	Bloomberg, CKAN, credit bureaus	Generates direct revenue
Indirect monetisation	Using data to improve products/services	Netflix, Google Ads, Prometheus, Flink	Drives growth, efficiency

Both monetisation strategies are important. Direct monetisation turns data into a commercial product. Indirect monetisation helps businesses operate efficiently. Together, they show that in today's economy, data is not just fuel — it's a product, a service and a strategy, increasingly powered by open source innovation.

Building data products: What one should know

Creating a data product requires more than basic programming and data collection activities. The process needs clear workflows, team collaboration and strong attention to data quality. A data product lifecycle has five essential stages.

The first stage is ingestion. The process of raw data collection occurs through APIs, mobile apps, IoT devices and logs. A food delivery application receives GPS information order data and rating feedback in real time. Open source tools like Apache NiFi help automate and monitor this process.

Next is cleaning. Teams correct errors, handle missing values and standardise data formats. The data processing tools Pandas, OpenRefine and dbt (data build tool) enable these operations. Machine learning projects that skip this step typically produce models with low accuracy.

The third stage is enrichment, where additional context is added. The process of integrating geocoding APIs with customer records enables location enrichment while integrating weather data with delivery times enables delay analysis. OpenWeatherMap and GeoNames serve as popular open data APIs used in enrichment pipelines.

Then comes storage. The cleaned data gets stored in systems such as PostgreSQL, Apache Hive or DuckDB for querying and analysis. The open source storage solutions MinIO and Apache Iceberg serve as popular choices for scalable storage needs. These tools allow data versioning alongside fast data access capabilities.

The final stage in the data product lifecycle is delivering the data. The data becomes available through dashboards, APIs and downloadable files. Data visualisation tools such as Apache Superset, Metabase and Grafana enable users to interpret information.

Multiple roles collaborate in building these data products. Data engineers create pipelines using tools like Apache Airflow. ML engineers train models using data from Jupyter notebooks or TensorFlow. Data analysts use JupyterLab, R or Plotly to explore trends and build reports. Product managers verify that the product addresses genuine user requirements such as time reduction and enhanced personalisation.

Some key practices are common across teams. Versioning with tools like DVC (Data Version Control) ensures traceability. Reproducibility ensures that running the same pipeline consistently yields the same result. This is essential for auditing. Observability using tools like GX (Great Expectations) or OpenTelemetry helps detect broken pipelines or outdated data.

Models of data valuation

The process of valuing data remains challenging. Data exists as a non-rivalrous resource because its value does not decrease when multiple users access it. In fact, the value of data increases when different datasets are combined. For example, merging sales data with user location information, browser history and app feedback creates a more valuable dataset. A college placement dashboard built with open source tools becomes significantly more effective after linking student scores, course history and recruiter feedback through a unified dataset that uses well-modelled data.

Organisations need to determine data value before making decisions about selling it, sharing it openly or using it internally. There are three primary models that determine data value — cost-based, market-based and value-based.

The cost-based model calculates value by adding how much it took to build the dataset — collection,

Figure 1: Stages of a data product lifecycle